

FORTINET
Training Institute

FortiGate Administrator

Intrusion Prevention and Application Control

FortiOS 7.6

Fortinet Certified Professional
Network Security

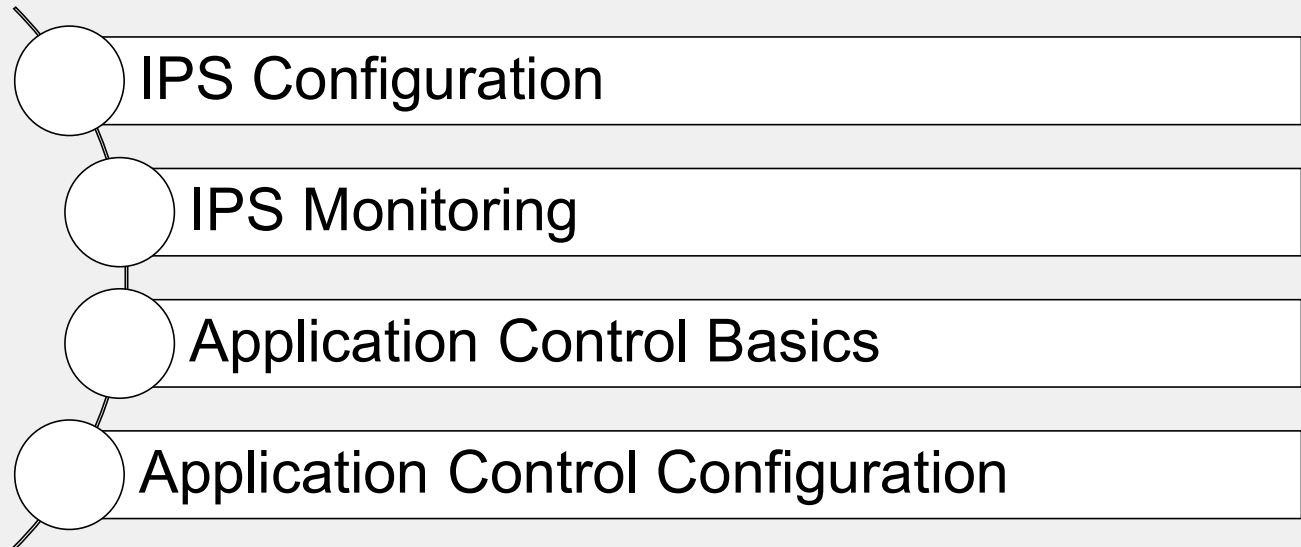
Fortinet Certified Professional
Security Operations

Fortinet Certified Professional
Public Cloud Security

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In this lesson, you will learn how to use FortiGate to protect your network against intrusions and monitor and control network applications that may use standard or nonstandard protocols and ports—beyond simply blocking or allowing a protocol, port number, or IP address.

Lesson Overview



In this lesson, you will learn about the topics shown on this slide.

IPS Configuration

Objectives

- Identify the different components of an intrusion prevention system (IPS) package
- Configure an IPS sensor

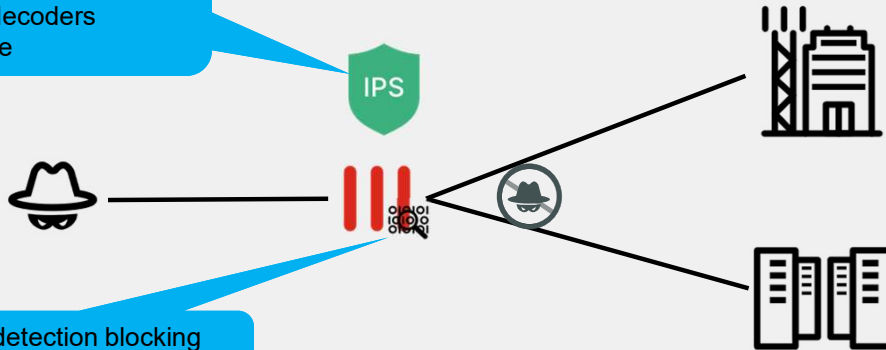
After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in IPS configuration, you will be able to implement an effective IPS solution.

IPS

IPS components include:

- IPS signature databases
- Protocol decoders
- IPS engine



Intrusion Prevention System (IPS) detects network attacks and prevents threats from compromising the network, including protected devices. IPS utilizes signatures, protocol decoders, heuristics (or behavioral monitoring), threat intelligence (such as FortiGuard Labs), and advanced threat detection can perform prevent exploitation of known and unknown zero-day threats. FortiGate IPS can perform deep packet inspection to scan encrypted payloads to detect and prevent threats from attackers.

IPS components include the following:

1. **IPS Signatures:** IPS on FortiGate uses signature databases to detect known attacks, like exploits. Rate-based IPS signatures also allows you to detect anomalies, which are unusual behaviors in the network, such as higher-than-usual CPU use or network traffic. Rate-based IPS signatures are part of behavioral analysis, like DoS policies and protocol constraints inspection, which detect and monitor (and, in some cases, block or mitigate) anomalies, because they reveal the symptoms of a new, never-previously-seen attack.
2. **Protocol decoders:** Protocol decoders can detect network errors and protocol anomalies. Protocol decoders parse each packet according to the protocol specifications. By understanding the protocol, the IPS can identify anomalies or malicious activities even if they are encapsulated in legitimate protocol behaviors. If the traffic doesn't conform to the specification—if, for example, it sends malformed or invalid commands to your servers—then the protocol decoder detects the error.
3. **IP Engine:** The IPS engine is responsible for IPS and protocol decoders, in addition to application control, flow-based antivirus protection, web filtering, and email filtering.

List of IPS Signatures

- Create new IPS sensors and view a list of predefined sensors

Security Profiles > Intrusion Prevention

Edit IPS Sensor

Name: default
Comments: Prevent critical attacks. 25/255
Block malicious URLs: ☐

IPS Signatures and Filters

[+ Create New](#) [Edit](#) [Delete](#)

Details	Exempt IPs	Action	Packet Logging
SEV: ■ ■ ■ ■ ■		Default	Disabled

Botnet C&C

Scan Outgoing Connections to Botnet Sites: Disable Block Monitor

View IPS Signatures

5,864 Total

Severity: High (2,530), Critical (1,029), Medium (1,765), Low (100), Information (61)

7,051 Total

Target: Server (3,685), Client (3,366)

OS: Windows (4,452), Linux (1,632), MacOS (1,072), All (445), Solaris (154), More...

[+ Create new](#) [Search](#)

IPS Signature	Name	Severity	Target	OS	Action	CVE-ID
☐ 3Com.3CDaemon.FTPServer.Buffer.Overflow	3Com.3CDaemon.FTPServer.Buffer.Overflow	High	Server	Windows	Block	CVE-2005-0277
☐ 3Com.3CDaemon.FTPServer.Information.Disclosure	3Com.3CDaemon.FTPServer.Information.Disclosure	Medium	Client	Windows	Pass	CVE-2005-0278
☐ 3Com.OfficeConnect.ADSL.Wireless.Firewall.Router.DoS	3Com.OfficeConnect.ADSL.Wireless.Firewall.Router.DoS	High	Server	Linux	Block	
☐ 427BB.Cookie.Based.Authentication.Bypass	427BB.Cookie.Based.Authentication.Bypass	High	Server	Other	Block	CVE-2006-0153
☐ A325.Botnet	A325.Botnet	High	Server/Client	All	Block	

FortiGate IPS sensor is a collection of IPS signatures and filters that define the scope of what the IPS engine will scan when the IPS sensor is applied. Multiple sets of signatures, filters, or both may be present in an IPS sensor. A set of IPS signatures consists of manually selected signatures, while a set of IPS filters consists of filters based on signature attributes like target, severity, protocol, OS, and application. Each signature has predefined attributes and an action, such as block, allow, monitor (pass), quarantine, and reset. It is also possible to create custom IPS signatures to apply to an IPS sensor.

FortiOS includes predefined IPS sensors with associated predefined signatures. Examples are default, all_default, protect_client, protect_http_server, and so on.

FortiGate downloads a FortiGuard IPS package and new signatures will appear in the signature list. When configuring FortiGate, you can change the **Action** setting for each sensor that uses a signature.

The default action setting is often correct, except in the following cases:

- Your software vendor releases a security patch. Continuing to scan for exploits wastes FortiGate resources.
- Your network has a custom application with traffic that inadvertently triggers an IPS signature. You can change the action setting until you notify Fortinet so that the FortiGuard team can modify the signature to avoid false positives.

Configuring IPS Sensors

- Add individual signatures
- Add groups of signatures using filters

Security Profiles > Intrusion Prevention

New IPS Sensor

Name:

Comments: 0/255

Block malicious URLs: ☐

IPS Signatures and Filters

+ Create New Edit Delete

Details Exempt IPs Action Packet Logging

No results

Add Signatures

Type: **Signature**

Action: **Default**

Packet logging: **Enable** **Disable**

Status: **Enable** **Disable** **Default**

Rate-based settings: **Default** **Specify**

Exempt IPs: 0 Edit IP Exemptions

Add All Results Search

Name	Severity	Target	OS	Action	CVE-ID
3Com.3CDAemon.FTPServer.Buffer.Overflow	High	Server	Windows	Block	CVE-2005-0277
3Com.3CDAemon.FTPServer.Information.Disclosure	Medium	Client	Windows	Pass	CVE-2005-0278
3Com.Intelligent.Management.Center.Information.Disclosure	High	Server	Windows	Block	

Add Signatures

Type: **Filter** **Signature**

Action: **Default**

Packet logging: **Enable** **Disable**

Status: **Enable** **Disable** **Default**

Filter: **Host** **Server** **IPv4** **IPv6** **Protocol** **HTTP** **OS** **Windows**

Add All Results Search

Name	Severity	Target	OS	Action	CVE-ID
Adobe.Acrobat.And.Reader.TrueTypeFont.Parsing.Buffer.Overflow	High	Server Client	All	Block	CVE-2012-0774
Adobe.Acrobat.BMP.Colors.Parsing.Memory.Corruption	High	Server Client	Windows MacOS	Block	CVE-2011-4373

There are two ways to add predefined signatures to an IPS sensor. One way is to select the signatures individually. After selecting a signature in the list, the signature is added to the sensor with its default action.

The second way to add a signature to a sensor is using filters. FortiGate adds all the signatures that match the filters.

The purpose of the IPS feature is to protect the inside of the network from outside threats.

Configuring IPS Sensors—Rate-Based Signatures

- Add rate-based signatures to block traffic when the threshold is exceeded during a time period

Security Profiles > Intrusion Prevention

Add Signatures

Type: Filter **Signature**

Action: Default

Packet logging: ☒ Enable ☒ Disable

Status: ☒ Enable ☒ Disable ☒ Default

Rate-based settings: Default **Specify**

Threshold: 0

Duration (seconds): 60

Track By: **Any** Source IP Destination IP

Exempt IPs: 0 Edit IP Exemptions

☒ Add All Results Search Selected

	Name	Severity	Target	OS	Action	CVE-ID
IPS Signature 5,864						
	3Com.3CDaemon.FTP.Server.Buffer.Overflow	High	Server	Windows	Block	CVE-2005-0277
	3Com.3CDaemon.FTP.Server.Information.Disclosure	Medium	Client	Windows	Pass	CVE-2005-0278
	3Com.Intelligent.Management.Center.Information.Disclosure	High	Server	Windows	Block	

These parameters are applicable to the signatures selected at the bottom

Can track the traffic based on source or destination IP address

You can also add rate-based signatures to block specific traffic when the threshold is exceeded. On the CLI, if you set the command `rate-mode` to `periodical`, FortiGate triggers the action when the threshold is reached during the configured **Duration** time period. You should apply rate-based signatures only to protocols you use. This saves system resources and can discourage a repeat attack. FortiGate does not track statistics for that client while it is temporarily blocklisted.

IPS Sensor Inspection Sequence

Security Profiles > Intrusion Prevention

New IPS Sensor

Name: Server IPS Profile

Comments: Write a comment... 0/255

Block malicious URLs: ☐

IPS Signatures and Filters

[+ Create New](#) [Edit](#) [Delete](#)

Details	Exempt IPs	Action	Packet Logging
Apache.Tomcat.Integer.Overflow.Information.Disclosure	0	Monitor	Disabled
IGT Server SEV SEV OS Windows		Default	Disabled

2

New entries are placed at the bottom of the list

IPS signatures and filters are processed in sequence

When the IPS engine compares traffic with the signatures in each filter, order matters. The rules are similar to firewall policy matching; the engine evaluates the filters and signatures at the top of the list first, and applies the first match. The engine skips subsequent filters.

So, position the most likely matching filters, or signatures, at the top of the list. Avoid making too many filters, because this increases evaluations and CPU usage. Also, avoid making very large signature groups in each filter, which increase RAM use.

In the event of a false-positive outbreak, you can add the triggered signature as an individual signature, and then set the action to **Monitor**. This allows you to monitor the signature events using IPS logs, while investigating the false-positive issue.

Configuring IP Exemptions

- Only configurable under individual IPS signatures

Security Profiles > Intrusion Prevention

IPS Signatures and Filters

+ Create New Edit Delete				
Details	Exempt IPs	Action	Packet Logging	
Apache.Tomcat.Integer.Overflow.Information.Disclosure	1	Monitor	Disabled	
TGT Server		Default	Disabled	
SEV				
SEV				
OS Windows				

Edit IP Exemptions

[+ Create New](#) [Delete](#)

Source IP/Netmask	Destination IP/Netmask
0.0.0.0/0	10.0.1.10/32

Exempt specific source or destination IP addresses

Sometimes, it is necessary to exempt specific source or destination IP addresses from specific signatures. This feature is useful during false-positive outbreaks. You can temporarily bypass affected endpoints until you investigate and correct the false-positive issue.

You can configure IP exemptions on individual signatures only. Each signature can have multiple exemptions.

IPS Actions

Security Profiles > Intrusion Prevention

Add Signatures

Type: Filter Signature

Action: Default

Packet logging: ☒ Allow, ☐ Monitor, ☐ Block, ☐ Reset, ☐ Default, ☐ Quarantine

Disables: Disable, Default

Search: [Q]

Signature	Sev...	Target	OS	Action	CVE-ID
IPS Signature 5,864					
HP.Database.Archiving.Software.GIOP.Parsing.Buffer...	5	Server	Windows Solaris	Block	CVE-2011-4164
Symantec.Gateway.Products.DNS.Cache.Poisoning	4	Client	Windows Solaris	Block	CVE-2005-0817
Oracle.Outside.In.OOXML.Tag.Parsing.Stack.Buffer.O...	5	Client	Windows Solaris	Block	
Oracle.Outside.In.Lotus123.Heap.Buffer.Overflow	4	Client	Windows Solaris	Block	CVE-2012-0110

When you create a new entry to add signatures or filters, you can select the action by clicking **Action**.

Select **Allow** to allow traffic to continue to its destination. Select **Monitor** to allow traffic to continue to its destination and log the activity. Select **Block** to silently drop traffic matching any of the signatures included in the entry. Select **Reset** to generate a TCP RST packet whenever the signature is triggered. Select **Default** to use the default action of the signatures.

Quarantine allows you to quarantine the attacker's IP address for a set duration. You can set the quarantine duration to any number of days, hours, or minutes.

When you enable **Packet logging**, FortiGate stores a local copy of the packet that matches the signature. This enhances the view of erroneous or suspicious packets. FortiGate's IPS packet logging helps to provide robust security operations by providing accurate information into network traffic, which are crucial for detecting, investigating, and mitigating cyber threats. Even though enabling packet logging consume significant storage, memory, and processing resources, you can enable packet logging for different use case scenarios such as detection and prevention of zero-day attacks, troubleshooting, behavioral analysis, threat hunting etc.

You can set these actions on hold for new FortiGuard IPS signature by enabling the `override-signature-hold-by-id` CLI command. During the time defined by the CLI command `signature-hold-time`, the action is then set to **Monitor** to avoid false positives, with a log created including the message 'signature is on hold'.

Enabling Botnet Protection

Security Profiles > Intrusion Prevention

Edit IPS Sensor

Name: Server IPS Profile

Comments: Write a comment... 0/255

Block malicious URLs: ☐

IPS Signatures and Filters

+ Create New Edit Delete

Details	Exempt IPs	Action	Packet Logging
Apache.Tomcat.Integer.Overflow.Information.Disclosure	0	Monitor	Disabled
TGT Server SEV ■ ■ ■ ■ ■ SEV ■ ■ ■ ■ ■ OS Windows		Default	Disabled

Botnet C&C

Scan Outgoing Connections to Botnet Sites: Disable Block Monitor

3100 IP Addresses in botnet package.

Set action to **Block** or **Monitor**

Botnet database from FortiGuard (included with a valid IPS license)

For consolidated botnet protection, you can enable botnet scanning on the IPS profile that you apply the firewall policy on.

There are three possible actions for **Botnet and C&C**:

- **Disable**: Do not scan connections to botnet servers
- **Block**: Block connections to botnet servers
- **Monitor**: Log connections to botnet servers

Applying IPS Inspection

Policy & Objects > Firewall Policy

Security Profiles

- AntiVirus ☐
- Web Filter ☐
- DNS Filter ☐
- Application Control ☐
- IPS ☒ **Enable IPS** Select the IPS security profile corresponding to the configured IPS sensors IPS protect_client
- File Filter ☐
- SSL Inspection ☒ Set deep-inspection for encrypted protocols SSL deep-inspection
- Decrypted Traffic Mirror ☐

Logging Options

Log Allowed Traffic ☒ **Enable logging** Security Events All Sessions

To apply an IPS sensor, you must enable **IPS** and then select the sensor in a firewall policy.

Certain vulnerabilities apply only to encrypted connections and FortiGate can't identify the threat reliably if it can't parse the payload. For this reason, you must use an SSL inspection profile, usually **deep-inspection**, if you want to get the maximum benefit from your IPS features.

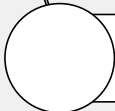
By default, FortiGate logs all security events. This means you can see any traffic that is being blocked or monitored by IPS.

If you think some traffic should be blocked but is passing through the policy, you should change the **Log Allowed Traffic** method to **All Sessions**. This logs all traffic processed by that firewall policy, and not just the traffic that is blocked or monitored by the security profiles. This can help you in identifying false negative events.

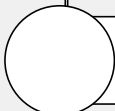
Lesson Progress



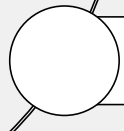
IPS Configuration



IPS Monitoring



Application Control Basics



Application Control Configuration

Good job! You now understand IPS on FortiGate.

Now, you will learn about IPS monitoring.

IPS Monitoring

Objectives

- Troubleshoot IPS high CPU usage
- Manage IPS fail open

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in troubleshooting, you should be able to identify, investigate, and manage some common issues with IPS deployments on FortiGate.

IPS Logging

Log & Report > Security Events

Summary Logs

35 Events

Intrusion Prevention

Top Attack

Attack	Action	Count
HTTP.URI.SQL.Injection	Dropped	6
NetworkActivWeb.Server.XSS	Dropped	4
Apache.Expect.Header.XSS	Detected	1
BadBlue.MFCIS.PICCommand.Remote.Buffer.Overflow	Detected	1

12 events

Summary Logs

Search

Intrusion Prevention

Disk

5 minutes

Details

Date/Time	Severity	Source	Protocol	User	Action	Count	Attack Name
2023/09/27 01:35:06	Medium	10.200.1.254	6		dropped		NetworkActivWeb.Serv...
2023/09/27 01:34:56	Medium	10.200.1.254	6		dropped		NetworkActivWeb.Serv...
2023/09/27 01:34:56	High	10.200.1.254	6		detected		PHPBB.Viewtopic.Highli...
2023/09/27 01:34:56	High	10.200.1.254	6		detected		PHPBB.Viewtopic.Highli...

Log Details

Action

dropped

Threat

16,384

Policy ID

2 (Web_Server-Access_IPS)

Policy UUID

fa88b646-5d0f-51ee-7fa0-894366c
dc738

Policy Type

Firewall

Security

Level

Alert Notification

Threat Level

Medium

Threat Score

10

Cellular

Service

HTTP

Intrusion Prevention

Profile

WEBSERVER

Attack Name

NetworkActivWeb.Server.XSS

Attack ID

34,971

Reference

<http://www.fortinet.com/ids/VID34971>

If you enabled security events logging in the firewall policies that apply IPS, the logs are available on the **Security Events** pane on the **Log & Report** page. You can view the logs by clicking on **Intrusion Prevention**.

You should review IPS logs frequently. The logs are an important source of information about the kinds of attacks that are being targeted at your network. This helps you develop action plans and focus on specific events, for example, patching a critical vulnerability.

Troubleshoot IPS High-CPU Usage

- CLI command to troubleshoot continuous high CPU use by IPS engines

```
# diag test application ipsmonitor <Integer>
```

IPS Engine Test Usage:

1: Display IPS engine information

2: Toggle IPS engine enable/disable status

5: Toggle bypass status

Shuts down IPS engine completely

99: Restart all IPS engines and monitor

IPS engine remains active, but does not inspect traffic

```
# diag test application ipsmonitor 1
```

```
pid = 2011, engine count = 1 (+1)
```

```
0 - pid:2058:2058 cfg:1 master:0 run:1
```

```
1 - pid:2832:2832 cfg:0 master:1 run:1
```

```
pid: 2832 index:1 master
```

```
version: 07006000FLEN07600-00007.00006
```

```
up time: 0 days 0 hours 5 minutes
```

```
init time: 0 seconds
```

```
socket size: 256(MB)
```

```
database: ipsetdb isdb
```

```
bypass: disable
```

While using IPS, short spikes in CPU usage by IPS processes can be caused by firewall policy or profile changes. These spikes are usually normal. Spikes might happen when FortiGate has hundreds of policies and profiles, or many virtual domains. Continuous high CPU use by the IPS engines is not normal, and you should investigate it. You can use the command shown on this slide, along with displayed options, to troubleshoot these issues.

If there are high CPU use problems caused by the IPS engine, you can use the `diagnose test application ipsmonitor` command with option 5 to isolate where the problem might be. Option 5 enables IPS bypass mode. In this mode, the IPS engine is still running, but it is not inspecting traffic. If the CPU use decreases after that, it usually indicates that the volume of traffic being inspected is too high for that FortiGate model.

If the CPU use remains high after you enable IPS bypass mode, it usually indicates a problem in the IPS engine, which you must report to Fortinet support. You can disable the IPS engine completely using option 2. If you want to restore IPS inspection of traffic after you finish troubleshooting, use option 2 again. At any time, you can check the status of the IPS engines using option 1.

Another recommendation to keep in mind is that if you need to restart the IPS, use option 99, as the slide shows. This guarantees that all the IPS-related processes restart correctly.

IPS Fail Open

- Fail open is triggered when the IPS socket buffer is full and new packets can't be added for inspection

```
config ips global
set fail-open <enable|disable>
...
end
```

Enable – bypass the packet without inspection
Disable – drop the new packet (default)

- IPS fail open entry log :

```
date=2024-11-4 time=09:07:59 logid=0100022700 type=event subtype=system
level=critical vd="root" logdesc="IPS session scan paused" action="drop"
msg="IPS session scan, enter fail open mode"
```

- When troubleshooting IPS fail-open events, try to identify a pattern
 - Has the traffic volume increased recently?
 - Does fail open trigger at specific times during the day?
- Create IPS profiles specifically for the traffic type
 - An IPS sensor configured to protect Windows servers doesn't need Linux signatures
 - Disable IPS on internal-to-internal policies

Default disable and packets are dropped!

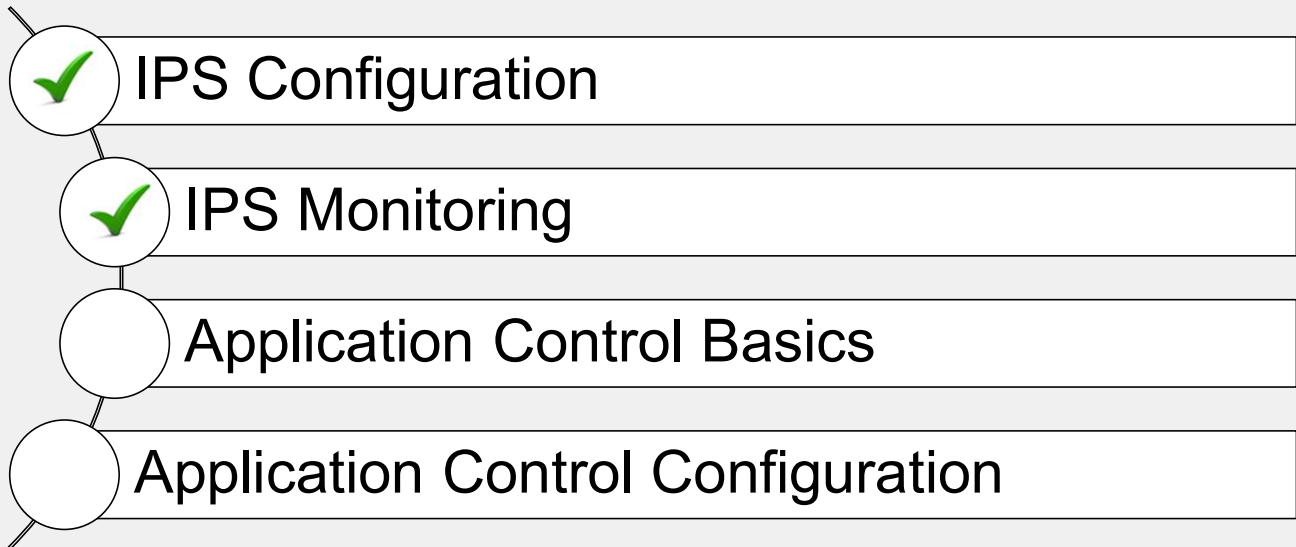
IPS goes into fail-open mode when IPS socket buffer is full for new packets. IPS engine has no space in memory to create more sessions and needs to decide whether to drop the sessions or bypass the sessions without inspection. What happens during this state depends on the IPS configuration. If the `fail-open` setting is enabled, some new packets (depending on the system load) will pass through without being inspected. By default, it is disabled, new packets are dropped.

IPS fail-open is recommended only in situations where network availability takes precedence over network security—such as in a datacenter.

What to identify in fail-open scenarios? Frequent IPS fail open events usually indicate that IPS can't keep up with the traffic demands. So, try to identify patterns. Has the traffic volume increased recently? Have throughput demands increased? Does fail open trigger at specific times during the day?

Once you identified the resource constraints try to tune and optimize your IPS configuration. Create IPS profiles specific to the type of traffic being inspected, and disable IPS profiles on policies that don't need them.

Lesson Progress



Good job! You now understand IPS monitoring on FortiGate.

Now, you will learn about application control basics.

Application Control Basics

Objectives

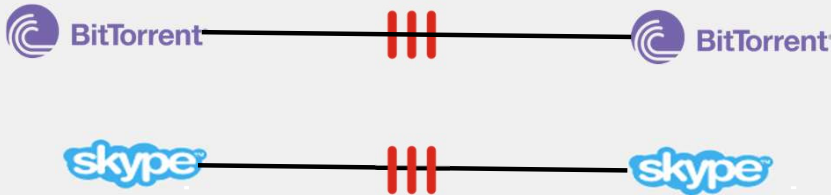
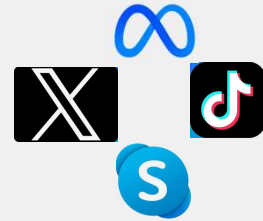
- Understand application control
- Use application control signatures

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in application control basics, you will be able to understand how application control works on FortiGate.

Application Control

- Uses the IPS engine in flow-based scan
- Detects and acts on network application traffic
- Appropriate for detecting peer-to-peer (P2P) applications

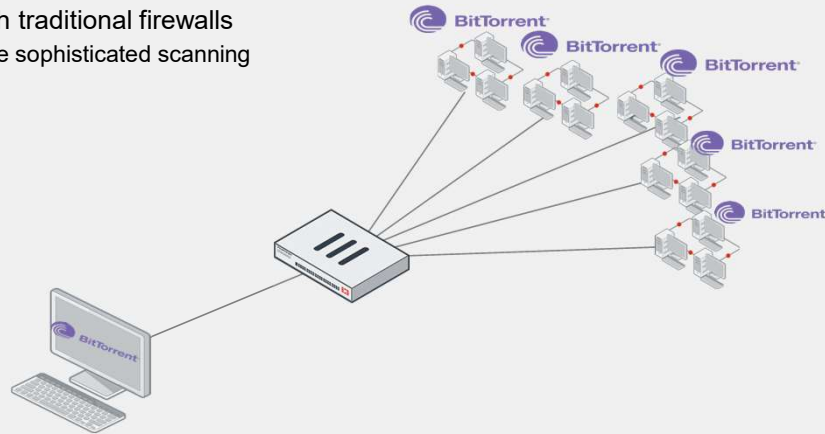


As previously mentioned, the IPS engine is also responsible for application control. You can configure application control in proxy-based and flow-based firewall policies. However, because application control uses the IPS engine, which uses flow-based inspection, the inspection is always flow-based.

Application control identifies applications, such as Google Talk, by matching known patterns to the application transmission patterns. Therefore, an application can be accurately identified, only if its transmission pattern is unique. However, not every application behaves in a unique way. Many applications reuse pre-existing, standard protocols and communication methods. For example, many video games, such as *World of Warcraft*, use the BitTorrent protocol to distribute game patches. Still, with the help of the IPS engine, application control analyzes network traffic and detects application traffic, even if the application is using standard or non-standard protocols and ports. It doesn't operate using built-in protocol states. As a consequence, application control is better suited for detecting P2P protocols, because they use port randomization, pinholes, and changing encryption pattern techniques.

Peer-to-Peer Architecture

- Peer-to-peer (P2P) download
 - One client
 - Many servers
 - Dynamic port numbers
 - Optionally, dynamic encryption
 - *Hard to block* with traditional firewalls
 - Requires more sophisticated scanning



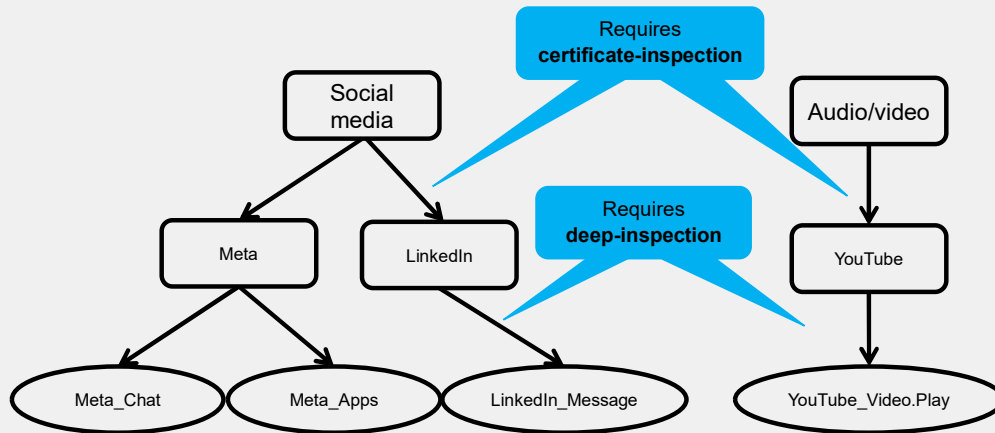
Peer-to-peer (P2P) downloads divide each file among multiple (theoretically unlimited) peers. Each peer delivers part of the file. While having many clients is a disadvantage in client-server architectures, it is an advantage for P2P architecture because, as the number of peers increases to n , the file is delivered n times faster.

Because popularity increases the speed of delivery—unlike traditional client-server architecture where popularity could effectively cause a denial of service (DoS) attack on the server—some software, such as BitTorrent distributions of Linux, and games distributing new patches, leverage this advantage. Even if each client has little bandwidth, together they can offer more bandwidth for the download than many powerful servers.

Consequently, in order to download the file, the requesting peer can consume much more bandwidth per second than it would from only a single server. Even if there is only one peer in your network, it can consume unusually large amounts of bandwidth. Because the protocols are usually evasive, and there will be many sessions to many peers, they are difficult to completely block.

Application Control—Hierarchical Structure

- Application control signatures are organized in a hierarchical structure
 - The parent signature takes precedence over the child signature



Many web applications offer functionality that can be embedded in third-party websites or applications. For example, you can embed a Meta **Like** button at the end of an article, or reference a YouTube video on an educational website. FortiOS gives administrators all the tools they need to inspect subapplication traffic. The FortiGuard application control signature database is organized in a hierarchical structure. This gives you the ability to inspect the traffic with more granularity. You can block Meta applications while allowing users to collaborate using Meta chat.

List of Application Signatures

Security Profiles > Application Control

New Application Sensor

93 Cloud Applications require deep inspection.
0 policies are using this profile.

Name

Comments 0/255

Categories

Mixed ▾ All Categories

- Business (179, ☁ 6)
- Collaboration (293, ☁ 6)
- Game (124)
- Mobile (3)
- P2P (85)
- Remote Access (91)
- Storage/Backup (296, ☁ 16)
- Video/Audio (206, ☁ 13)
- Web Client (18)
- Cloud/IT (31)
- Email (87, ☁ 12)
- General Interest (241, ☁ 9)
- Network Service (332)
- Proxy (106)
- Social Media (150, ☁ 31)
- Update (48)
- VoIP (31)
- Unknown Application

Firmware & General Updates License
Licensed (Expiration Date: 2026/01/17)

Application Control Signatures Package
Version 6.00741

Application Signatures
[View Application Signatures](#)

Filter option

Application signatures

Q linkedin

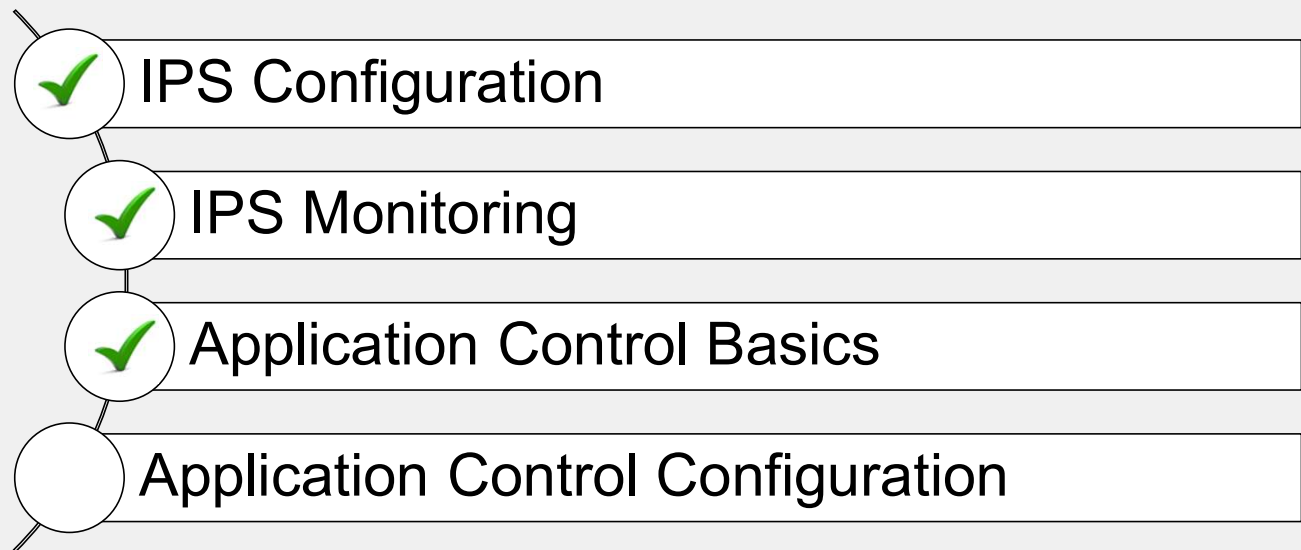
	Name	Category	Popularity	Risk
☐	Application signature 7/2414			
☐	LinkedIn	Social Media	★★★★★	■ □ □ □ □
☐	LinkedIn_Apps	Social Media	★☆☆☆☆	■ □ □ □ □
☐	LinkedIn_File.Download	Social Media	★★★★★	■ □ □ □ □
☐	LinkedIn_File.Upload	Social Media	★★★★★	■ □ □ □ □
☐	LinkedIn_Login	Social Media	★★★★★	■ □ □ □ □
☐	LinkedIn_Mail	Social Media	★★★★★	■ □ □ □ □
☐	LinkedIn_Post	Social Media	★★★★★	■ □ □ □ □

Active signature database

After FortiGate downloads a FortiGuard Application Control Signature package, new signatures appear in the signature list.

In the example shown on this slide, the signatures are filtered with `LinkedIn`, showing its category and the corresponding hierarchical structure.

Lesson Progress



Good job! You now understand basic application control functionality.

Now, you will learn about application control configuration.

Application Control Configuration

Objectives

- Configure and apply application control profile
- Monitor application control events
- Troubleshoot traffic matching with application control profile issues

After completing this section, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in application control feature configuration and monitoring, you will be able to use and maintain application control in profile mode.

Filters Actions

Security Profiles > Application Control

Video/Audio (206, 13)

- Monitor**: Allows and also logs
- Allow**: Continues to next scan or feature and does not log
- Block**: Drops packets and logs
- Quarantine**: Blocks and log traffic from the suspected source until the expiration time
- View Signatures (206)**: View the list of signatures of native or cloud applications for a specific category
- View Cloud Signatures (13)**: View the list of signatures of native or cloud applications for a specific category

For each filter in the application control profile, you must indicate an action—what FortiGate does when traffic matches. Actions include the following:

- **Allow**: Passes the traffic and does not generate a log
- **Monitor**: Passes the traffic, but also generates a log message
- **Block**: Drops the detected traffic and generates a log message
- **Quarantine**: Blocks the traffic from the suspected source until the expiration time is reached, and generates a log message

The **View Signature** action allows you to view signatures from a particular category only and is *not* a configurable action. The **View Cloud Signatures** action allows you to view application signatures for cloud applications from a particular category.

Which is the correct action to choose?

If you're not sure which action to choose, **Monitor** can be useful initially, while you study your network. Later, after you have studied your network traffic, you can fine-tune your filter selection by choosing the most appropriate action. The action you choose also depends on the application. If an application requires feedback to prevent instability or other unwanted behavior, then you might choose **Quarantine** instead of **Block**. Otherwise, the most efficient use of FortiGate resources is to block.

Configuring Additional Options

The screenshot shows the 'Security Profiles > Application Control' configuration page. A red box highlights the 'Network Protocol Enforcement' toggle, which is turned on. A blue callout points to this toggle with the text: 'Allows blocking or monitoring of known services on unknown ports'.

Below this, there are sections for 'Create New', 'Enforce Protocols', and 'Violation Action'. A red box highlights the 'Create New' button. A blue callout points to it with the text: 'Allows blocking or monitoring of known services on unknown ports'.

The 'Enforce Protocols' section shows a table with columns for Port, Enforce Protocols, and Violation Action. A red box highlights the 'Enforce Protocols' column. A blue callout points to it with the text: 'Allows blocking or monitoring of known services on unknown ports'.

The 'Violation Action' section shows a table with columns for Priority, Details, Type, and Action. A red box highlights the 'Type' column. A blue callout points to it with the text: 'Allows blocking or monitoring of known services on unknown ports'.

The 'Options' section at the bottom has three items: 'Block applications detected on non-default ports' (checked), 'Allow and Log DNS Traffic' (unchecked), and 'Replacement Messages for HTTP-based Applications' (checked). Red boxes highlight each of these three options. Blue callouts point to them with the following text:

- 'Block applications detected on non-default ports': 'Forces applications to run on its default port'
- 'Replacement Messages for HTTP-based Applications': 'Applies only to HTTP/HTTPS applications'

On the right side, a 'New Default Network Service' dialog is open. It shows a list of 'Select Entries' on the right, including DNS, FTP, HTTP, HTTPS, IMAP, NNTP, POP3, SMTP, SNMP, SSH, and TELNET. A red box highlights this list. A blue callout points to it with the text: 'List of known services'.

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The **application sensor** provides also additional options.

Enabling the **Network Protocol enforcement** option allows you to configure network services (for example, FTP, HTTP, and HTTPS) on known ports (for example, 21, 80, and 443), while blocking those services on other ports.

This feature takes action in the following scenarios:

- When one protocol dissector confirms the service of network traffic, **Network Protocol Enforcement** can check whether the confirmed service is allowlisted under the server port. If it is not, then the traffic is considered a violation and IPS can take the action (for example, block) specified in the configuration.
- When there is no confirmed service for network traffic. It would be considered a service violation if IPS dissectors rule out all the services enforced under its server port, for example, if port 21 is configured for FTP and the protocol dissector could not decide on the exact service but is sure it is not FTP. If the port of the non-FTP traffic is 21, it will be a violation.

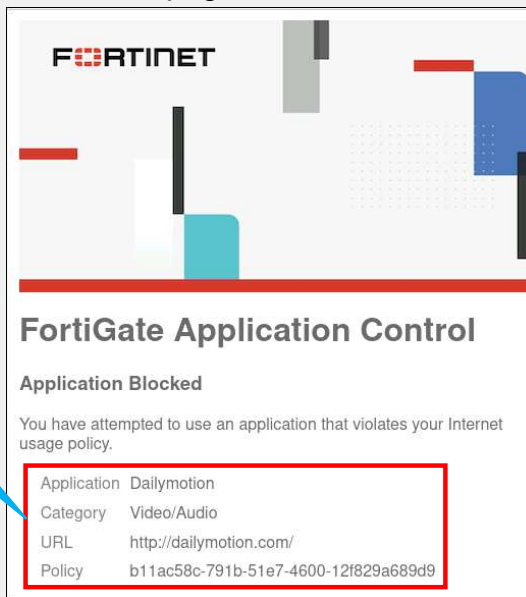
When the **Block applications detected on non-default ports** option enabled, FortiGate compares the ports used by the application with the ones defined in FortiGuard application signatures. The traffic is blocked if it does not match.

The **Replacement Messages for HTTP-based Applications** setting allows you to replace blocked content from HTTP/HTTPS applications with an explanation for the user's benefit. For non-HTTP/HTTPS applications, FortiGate only drops the packets or resets the TCP connection.

HTTP Block Page

- Application control HTTP block page

Information related to the HTTP page being blocked



You can customize the replacement message in **System > Replacement Messages**

For HTTP-based applications, application control can provide feedback to the user about why their application was blocked. This is called a block page, and it is similar to the one you can configure for URLs that you block using FortiGuard web filtering.

It is also worth mentioning that, if deep inspection is enabled in the firewall policy, all HTTPS-based applications provide this block page.

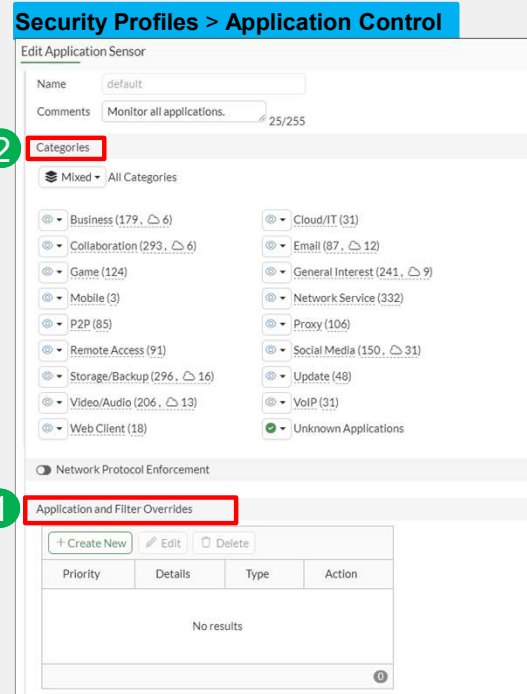
The block page contains the following information:

- Signature that detected the application (in this case, Dailymotion)
- Signature category (in this case, Video/Audio)
- URL that was specifically blocked (in this case, the index page of www.dailymotion.com), since a web page can be assembled from multiple URLs
- User name (if authentication is enabled)
- Group name (if authentication is enabled)
- Universal unique identifier (UUID) of the policy governing the traffic

The last item in this list can help you to identify which policy on FortiGate blocked the page, even if you have a large number of policies with many FortiGate devices securing different segments.

Scanning Order

- The IPS engine identifies the application
- The application control profile scans for matches in this order:
 1. Application and filter overrides
 2. Categories



With these multiple filters, which one has the priority?

After the IPS engine examines the traffic stream for a signature match, FortiGate scans packets for matches, in this order, for the application control profile:

1. Application and filter overrides: If you have configured any application overrides or filter overrides, the application control profile considers those first. It looks for a matching override starting at the top of the list, like firewall policies.
2. Categories: Finally, the application control profile applies the action that you have configured for applications in your selected categories.

Order of Scan and Blocking Behavior (Scenario 1)

Security Profiles > Application Control

Name: default
Comments: Monitor all applications. 25/255

Categories

- Mixed ▾ All Categories
- Business (179, 6)
- Collaboration (293, 6)
- Game (124) **Block**
- Mobile (3)
- P2P (85)
- Remote Access (91)
- Storage/Backup (296, 16)
- Web Client (18)
- Cloud/IT (31)
- Email (87, 12)
- General Interest (241, 9)
- Network Service (332)
- Proxy (106)
- Social Media (150, 31)
- Unknown Applications
- VoIP (31)

Network Protocol Enforcement

Application and Filter Overrides

Priority	Details	Type	Action
1	Battle.Net Dailymotion	Application	Monitor
2	BNR Excessive-Bandwidth	Filter	Block

Application overrides set for Battle.Net and Dailymotion applications

Filter overrides set for applications that consume excessive bandwidth

The Game and Video/Audio categories are set to Block and all other categories are set to Monitor

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In the example profile shown on this slide, the application control profile blocks the **Game** and **Video/Audio** categories. All other categories are set to **Monitor**, except **Unknown Applications**, which is set to **Allow**.

In the **Application and Filter Overrides** section, you can see that some exceptions are specified. Instead of being set to **Block**, **Battle.Net (Game)**, and **Dailymotion (Video/Audio)** are set to **Monitor**. Because application overrides are applied first in the scan, these two applications are allowed and generate logs.

Next, the scan checks for **application and filter overrides**. Because a filter override is configured to block applications that use excessive bandwidth, it blocks all applications using excessive bandwidth, regardless of other categories that allow these applications.

This slide shows an example of how several security profile features could work together, overlap, or work as substitutes, on the same traffic.

After FortiGate completes the application control profile scan, it begins other scans, such as web filtering. The web filtering scan could block Battle.Net and Dailymotion, but it would use its own block message. Also, web filtering doesn't check the list of application control overrides. So, even if an application control override allows an application, web filtering could still block it.

Similarly, static URL filtering has its own exempt action, which bypasses all subsequent security checks. However, application control occurs before web filtering, so that the web filtering exemption *cannot* bypass application control.

Order of Scan and Blocking Behavior (Scenario 2)

Security Profiles > Application Control

Name: default
Comments: Monitor all applications. 25/255

Categories

Mixed ▾ All Categories

- Business (179, ☁ 6)
- Collaboration (293, ☁ 6)
- Game (124)**
- Mobile (3)
- P2P (85)
- Remote Access (91)
- Storage/Backup (296, ☁ 16)
- Video/Audio (206, ☁ 13)**
- Web Client (18)
- Cloud/IT (31)
- Email (87, ☁ 12)
- General Interest (241, ☁ 9)
- Network Service (332)
- Proxy (106)
- Social Media (150, ☁ 31)
- Update (48)
- VoIP (31)
- Unknown Applications

☐ Network Protocol Enforcement

Application and Filter Overrides

+ Create New Edit Delete

Priority	Details	Type	Action
1	Excessive-Bandwidth	Filter	Block
2	Battle.Net Dailymotion	Application	Monitor

The filter override entry is moved above the application override entry

1 2 3

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In the example profile shown on this slide, the filter override has been moved above the application override. In this scenario, the filter override (**Excessive-Bandwidth**) is blocked and, since Dailymotion falls under the excessive bandwidth category, Dailymotion is blocked even though it is set to **Monitor** in the **Application and Filter Overrides** section.

The priority in which application and filter overrides are placed takes precedence.

Applying an Application Control Profile

- You must apply the **application control** profile on a firewall policy to scan the passing traffic
- Enable logging for security events or all sessions to log application control events.

Policy & Objects > Firewall Policy

Security Profiles

AntiVirus	☐
Web Filter	☐
DNS Filter	☐
Application Control	☒ APP default
IPS	☐
File Filter	☐
SSL Inspection	☒ SSL deep-inspection
Decrypted Traffic Mirror	☐

Logging Options

Log Allowed Traffic	☒ Security Events All Sessions
---------------------	--

Enable Application Control and select the profile

Use deep-inspection profile to scan encrypted traffic

Enable logging

After you configure an application control profile, you must apply it to a firewall policy. This instructs FortiGate to start scanning application traffic that is subject to the firewall policy.

When you enable the logging of security events or all sessions on a firewall policy, application control events are also logged. It allows you to monitor the application control use.

Monitoring Application Control Logging

Log & Report > Security Events

Summary Logs

111 Events

Application Control

Top Category	Action	Count
Web.Client	Pass	70
General.Interest	Pass	25
Video/Audio	Block	8
Network.Service	Pass	5

100 events

Logs

Date/Time4-10-07 16:22:04 Search

Date/Time	Source	Destination	Application Name	Action
2024/10/07 16:22:04	10.0.1.10	3.134.235.187	Dailymotion	Block
2024/10/07 16:22:04	10.0.1.10	3.134.235.187	HTTPS.BROWSER	Pass
2024/10/07 16:22:04	10.0.1.10	3.134.235.187	Dailymotion	Block
2024/10/07 16:22:04	10.0.1.10	3.134.235.187	HTTPS.BROWSER	Pass
2024/10/07 16:22:00	10.0.1.10	3.134.235.187	HTTPS.BROWSER	Pass
2024/10/07 16:21:58	10.0.1.10	142.251.111.94	Google.Accounts	Pass

Log Details

Application Control

Sensor: default

Application Name: Dailymotion

Application ID: 16072

Category: Video/Audio

Application Risk: Low

Protocol: 6

Service: HTTP

Message: Video/Audio: Dailymotion

Action

Action: Block

Policy ID: Application_Control (1)

Policy UUID: b11ac58c-791b-51e7-4600-12f829a689d9

Policy Type: Firewall

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FortiGate logs all application control events on the **Log & Report > Security Events** page. You can view the logs by clicking **Application Control**.

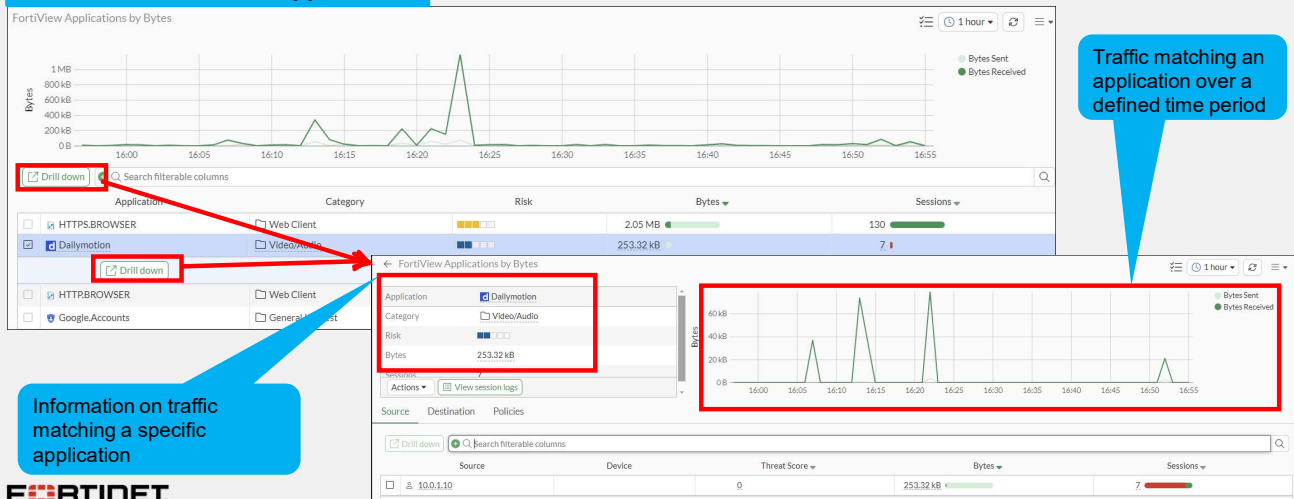
In the example shown on this slide, the default application control profile blocks access to Dailymotion. You can view this information in the **Log Details** section, as well as information about the log source, destination, application, and action.

You can also view the details on the **Forward Traffic** logs page, where firewall policies record activity. You can also find a summary of the traffic to which FortiGate applied application control. Again, this is because application control is applied by a firewall policy. To find out which policy applied application control, you can review either the **Policy ID** or the **Policy UUID** fields of the log message.

Troubleshoot Traffic Matching Application Control Profile

- Apply application control only to the traffic that requires it, and enable logging
- Review the logs and modify the configuration according to observations

Dashboard > FortiView Applications



Because not all traffic requires an application control scan, you must monitor the security event logs. If a traffic match is incorrect, you must then modify your configuration by first finding the firewall policy involved. This firewall policy reference is available in the **security events** logs and **forward traffic** logs.

You can also check the traffic matching with application control profiles on the **Dashboard > FortiView Applications** page. You can then select a specific application and drill down to view the sessions and bytes information for the traffic matching that application.

Knowledge Check

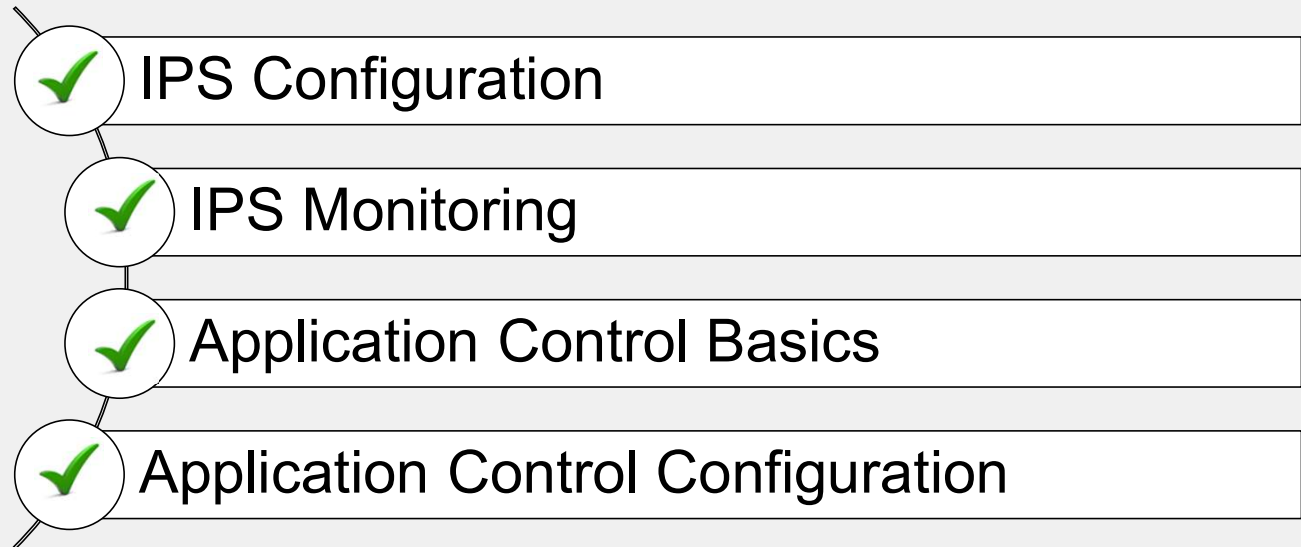
1. Which IPS component is responsible for application control?

- A. Protocol decoders
- ✓ B. IPS engine

2. Which statement about the HTTP block page for application control is true?

- ✓ A. It can be used only for web applications.
- B. It works for all types of applications.

Lesson Progress



Congratulations! You have completed this lesson.

Now, you will review the objectives covered in this lesson.

Review

- ✓ Configure an IPS sensor
- ✓ Troubleshoot IPS high CPU usage and IPS fail open
- ✓ Configure and apply an application control profile
- ✓ Monitor application control events
- ✓ Troubleshoot traffic matching with application control profile issues

This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned how to configure, maintain, and troubleshoot the FortiGate IPS solution. You also learned how to use methods beyond simply blocking protocols, port numbers, or IP addresses, to monitor and control both standard and non-standard network applications.